

Paper 1B drill: uncertainties

Seventeen short items. The arithmetic that loses marks.

IB Physics (2023 syllabus)

with worked answers

Name: _____

Date: _____

Total: **24 marks**

Three rules cover almost everything here. For a *sum or a difference*, add the absolute uncertainties. For a *product or a quotient*, add the percentage uncertainties. For a *power*, multiply the percentage uncertainty by the power.

A *absolute and percentage uncertainty*

- A1. A length is $2.40 \text{ m} \pm 0.05 \text{ m}$. Find its percentage uncertainty. [1]
- A2. A mass of 250 g is quoted as $\pm 2\%$. Find its absolute uncertainty. [1]
- A3. A time is $4.62 \text{ s} \pm 0.01 \text{ s}$. Find its percentage uncertainty. [1]

B *quoting a result to matching precision*

- B1. A student writes $d = 3.4567 \text{ cm} \pm 0.1 \text{ cm}$. Rewrite this properly. [1]
- B2. A resistor is $47 \Omega \pm 0.5\%$. Express it with an absolute uncertainty. [2]
- B3. A calculator gives $g = 9.81234 \text{ m s}^{-2}$ from an experiment whose uncertainty is $\pm 0.4 \text{ m s}^{-2}$. Quote the result. [1]

C *sums and differences*

- C1. Two masses, $120 \pm 2 \text{ g}$ and $85 \pm 3 \text{ g}$, are combined. State the total with its uncertainty. [1]
- C2. A liquid cools from $68.0 \pm 0.5^\circ\text{C}$ to $24.0 \pm 0.5^\circ\text{C}$. State the temperature drop with its uncertainty. [2]
- C3. A rod is $154.2 \pm 0.2 \text{ mm}$ long and a slot in it is $151.8 \pm 0.2 \text{ mm}$ long. Find the difference, its absolute uncertainty, and its *percentage* uncertainty. Comment on what has happened. [3]

D *products and quotients*

- D1. A rectangle is $12.0 \pm 0.1 \text{ cm}$ by $8.0 \pm 0.1 \text{ cm}$. Find its area with an absolute uncertainty. [3]
- D2. A trolley covers $100.0 \pm 0.5 \text{ m}$ in $12.4 \pm 0.1 \text{ s}$. Find its speed with an absolute uncertainty. [3]

E *powers and roots*

- E1. A sphere has radius 5.0 ± 0.1 cm. Find the percentage uncertainty in its volume. [2]
- E2. A period is 1.42 ± 0.02 s. Find the percentage uncertainty in T^2 . [1]
- E3. A pendulum's length is known to 3.0%. Given $T \propto \sqrt{L}$, find the percentage uncertainty in T . [1]

F *which term is worth fixing*

A specific heat capacity is found from $c = \frac{VIt}{m\Delta T}$, with these percentage uncertainties: V 0.85%, I 1.30%, t 0.33%, m 0.10%, ΔT 7.04%.

- F1. Find the percentage uncertainty in c . [1]
- F2. A classmate buys a better ammeter and voltmeter, making the uncertainties in V and I negligible. Find the new percentage uncertainty in c . [1]
- F3. Instead, the temperature rise is measured twice as precisely, so the ΔT term halves. Find the new percentage uncertainty in c , and state which purchase was the better one. [2]

Answers

A *absolute and percentage*

A1. $\frac{0.05}{2.40} \times 100 = 2.1\%$.

A2. 2% of 250 g is ± 5 g.

A3. $\frac{0.01}{4.62} \times 100 = 0.22\%$.

B *matching precision*

B1. $d = 3.5 \text{ cm} \pm 0.1 \text{ cm}$. The uncertainty is in the first decimal place, so quoting four decimals in the value claims a precision the measurement does not have.

B2. 0.5% of 47 is 0.235, which rounds to 0.2Ω . So $R = 47.0 \Omega \pm 0.2 \Omega$. Note the value gains a decimal place so that it matches the uncertainty.

B3. $g = 9.8 \pm 0.4 \text{ m s}^{-2}$. An uncertainty of 0.4 makes every digit after the first decimal meaningless.

C *sums and differences: add the absolute uncertainties*

C1. $120 + 85 = 205 \text{ g}$ and $2 + 3 = 5 \text{ g}$, so $205 \pm 5 \text{ g}$.

C2. $68.0 - 24.0 = 44.0^\circ\text{C}$, and the absolute uncertainties still *add* even though the readings are subtracted: $0.5 + 0.5 = 1.0$. So $44.0 \pm 1.0^\circ\text{C}$.

C3. $154.2 - 151.8 = 2.4 \text{ mm}$, uncertainty $0.2 + 0.2 = 0.4 \text{ mm}$, so $2.4 \pm 0.4 \text{ mm}$, which is $\frac{0.4}{2.4} \times 100 = 17\%$. Each length was measured to about 0.13%, yet their difference is known only to 17%, more than a hundred times worse. The absolute uncertainty stayed small; it is the *quantity* that shrank, and percentage uncertainty is a ratio. Subtracting two nearly equal numbers is the classic way to destroy the precision of a good measurement, and it is worth spotting in a method before you do the experiment rather than after.

D *products and quotients: add the percentage uncertainties*

D1. $A = 12.0 \times 8.0 = 96 \text{ cm}^2$. Percentages: $0.1/12.0 = 0.83\%$ and $0.1/8.0 = 1.25\%$, adding to 2.08%. Then 2.08% of 96 is 2, so $A = 96 \pm 2 \text{ cm}^2$.

D2. $v = 100.0/12.4 = 8.06 \text{ m s}^{-1}$. Percentages: $0.5/100.0 = 0.50\%$ and $0.1/12.4 = 0.81\%$, adding to 1.31%. Then 1.31% of 8.06 is 0.11, so $v = 8.06 \pm 0.11 \text{ m s}^{-1}$.

E *powers: multiply the percentage by the power*

- E1. $0.1/5.0 = 2\%$, and $V \propto r^3$, so $3 \times 2\% = 6\%$.
- E2. $0.02/1.42 = 1.41\%$, and squaring doubles it: $2 \times 1.41\% = 2.8\%$.
- E3. A square root is the power $\frac{1}{2}$, so $\frac{1}{2} \times 3.0\% = 1.5\%$. Taking a root *improves* the percentage uncertainty.
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F *which term is worth fixing*

- F1. $0.85 + 1.30 + 0.33 + 0.10 + 7.04 = 9.6\%$.
- F2. Removing V and I entirely leaves $0.33 + 0.10 + 7.04 = 7.5\%$. Two new instruments bought barely two percentage points.
- F3. Halving only the ΔT term gives $0.85 + 1.30 + 0.33 + 0.10 + 3.52 = 6.1\%$, a bigger improvement than perfect meters would have given. The better purchase is the one that attacks ΔT , because at 7.04% of a 9.6% total it is nearly three quarters of the whole budget. The general lesson: improve the *largest* term first, and check the numbers before spending, because the term that feels inaccurate is often not the one that matters.